

Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant $c_3 = \frac{1}{8\pi}$ and a five-slot carrier — build E_8 and read off the Standard Model: status assessment and reading guide

June 12, 2026

In one sentence

Topological Fixed-Point Theory (TFPT) derives the discrete structure of the Standard Model — gauge group, three families, hypercharges, the flavor skeleton — and its dimensionless constants from exactly **two inputs**: a boundary constant $c_3 = \frac{1}{8\pi}$ and a five-slot carrier $g_{\text{car}} = 5$. The two inputs build the exceptional lattice E_8 as a consistency checksum ($D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$), and the physics is read off as *fixed points*: the fine-structure constant $\alpha^{-1} = 137.0359992$, the mass ladder, the mixing angles and the scale grammar — every load-bearing claim machine-checked.

The TFPT document set — what is treated where

Plain language: TFPT is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard Model, the constants and the scale grammar. The development is **eight documents**, best read in order (this is the entry point; a detailed map is in §1):

1. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and proof ledger.
2. **tfpt_1_architecture_e8** — the two axioms, the derivation map, the EM fixed point α^{-1} , the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
3. **tfpt_2_standard_model** — the SM in one φ_0 -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures, and gravity/QG as the seam response.
4. **tfpt_3_e8_audit_bootstrap** — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. **tfpt_4_frontier** — honest status of η_B , m_p/m_e , Koide, dark matter, full QG.
- R. **tfpt_research_contracts** — the **open gates as numbered contracts**: lemma chains, closing theorems and machine-certifiability for (U_{wall}) and $G_{\text{metric}}/G_{\text{net}}$, with the current reduction state (selector triangle, index-4 identification).
- H. **tfpt_horizon_readouts** — **Appendix H** (unit-system reframe, *not* new physics): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock.
- O. **origin_theory** — **Origin Theory**: why no free fundamental number remains. The $(g_{\text{car}}, N_{\text{fam}})=(5, 3)$ skeleton, the triply-forced 8 (geometry = lattice = gravity), the order-30 Coxeter cycle, one boundary transport for flavor and horizon, and the gapped *unique attractor*. [I]/[L] core (v53–v56) plus one honestly-typed [P] cyclic interpretation.

You are reading document #1.

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Reading the markers (refined). The former overloaded “[M]” is split into the grade it actually carries, so a reviewer can see at a glance which claims are exact, which are formal, and which are conditional: **[I]** *exact identity* (e.g. $240 = 16 \cdot 5 \cdot 3$); **[L]** *Lie/lattice theorem* (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$); **[F]** *formalised* (Lean- or script-verified); **[N]** *numerical fixed point* (reproduced, stable); **[P]** *physical/conditional* (follows under named hypotheses); **[A]** *axiom or open*.

Physical — under hypotheses	RG masses, QG QG.AMB.01
Numerical — fit-free fixed point	$\alpha^{-1}, \Lambda, \theta_{12}$ EM.FP.01
Lattice — Lie / glue / root system	$E_8 = (D_5 \oplus A_3) + \mu_4$ E8.GLU.01
Formal — Lean / exact script	P2 algebra FORM.P2.02
Axiom — declared inputs	P1 c_3 , P2 g_{car} AX.P1.01

The claim stack (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv`.

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* **[I]** (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1} - p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L) ; (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{1, a\}$; (v) a Plücker coordinate of the anchor plane $\langle 1, a \rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* (“look, another number”) into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support)

1. the architecture: the two axioms $\{c_3, g_{\text{car}}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8 = (D_5 \oplus A_3) + \mu_4$ — document 1, Part II, machine-checked

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(verification/v1_e8_glue.py);
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3. the α^{-1} fixed point plus ablation (v3_em_alpha.py);
4. the **single status ledger** verification/status_ledger.csv — the source of truth for every claim’s grade, location, dependency and script;
5. the **open gates** as numbered research contracts (tfpt_research_contracts: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

1 What this is about: the jump in construction principle

In plain words — what we found, and why it matters

What we found. TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers (2, 3, 5, 16, ...) reappears in every sector: those numbers are not separate lucky hits. A single tiny “machine” — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8 , and E_8 is the unimodular “audit hull” that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not “printed” for free — where a scheme or analytic interface is needed, it is named).

Why this is simple. There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

How plausible it is. A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune*. The only genuinely free, continuous ingredient that remains is the number π .

Sharper still: one anchor $a = (1, 1, 2)$, and E_8 as a *compiler hull* [L]/

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|, \quad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$ generate the big Lie data directly: $p_1=4=|\mu_4|$, $p_2=6=|R^+(A_3)|$, $p_3=10=\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \text{rank } E_8,$$

with the binary ladder $p_{n+1} - p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\text{car}}\}$ to $a = (1, 1, 2)$ plus π . [verification/v23_anchor_generator.py]

And a is itself *half-forced* ([L]/[P]): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\text{fam}} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2 + 1 + 1$ is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not “why $a = (1, 1, 2)$ ” but “why the carrier $g_{\text{car}} = 5 / (|\mu_4|, N_{\text{fam}}) = (4, 3)$ ” — that interface (with c_3) is the [A] input layer.

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not “everything in the adjoint”. This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor $a = (1, 1, 2)$ into the E_8 audit hull.*

In one line:

The anchor generates the syntax; the boundary generates the scale; E_8 checks the consistency.

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0 , α^{-1} , the exponential scales) flows from π ; and E_8 is the unimodular *checksum container*, not the world group.

The anchor ratio triad ([[verification/v119_review_validation_2.py](#)]). The three elementary symmetric values, normalised by the family count, are the theory’s three recurring fractions [[I](#)]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio), } \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain), } \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow’s critical points are exactly $(2, 5) = (e_3, e_2)$ with inflection $\frac{7}{2} = \frac{e_3+e_2}{2}$: the fractions $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [[I](#)]: $\Omega_{\text{adm}} = 2p_1p_2 = 48$, $10b_1 = 1 + p_1p_3 = 41$, $\Delta_Y = e_2^2 = p_0^2 + 2 \text{ rank } E_8 = 25$, $h(E_8) = e_3p_0e_2 = 30$, $\text{rank } E_8 = p_0 + e_2$.

That is now the case. The whole development is consolidated into **seven companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

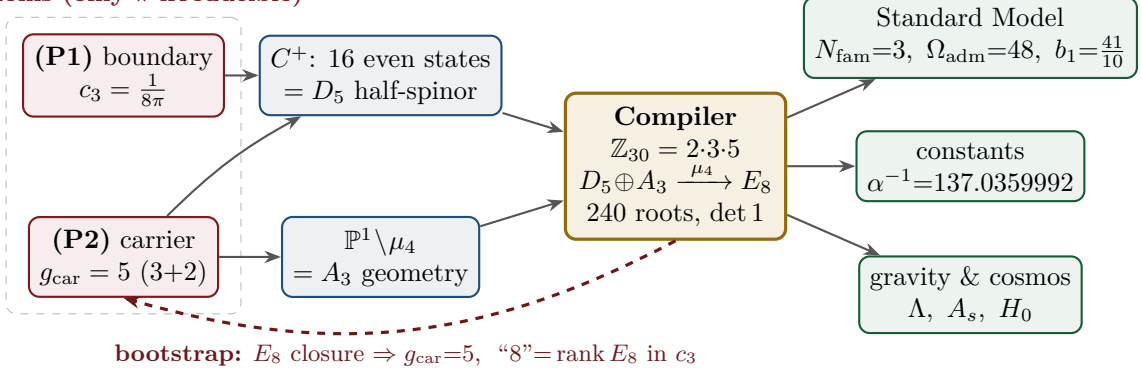
Document	What it contains (former notes now as parts)
tfpt_1_architecture_e8	Core. The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}=2\cdot3\cdot5$, the explicit $D_5 \oplus A_3 + \mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\text{Spin}(10) \times \text{SU}(4)) / \Delta \mathbb{Z}_4$ (Part II).
tfpt_2_standard_model	Standard Model. The full SM in one φ_0^{ret} -ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/ n_s, r , Starobinsky M , derived θ_{12} , quark c , the H2 splitting type, Λ -typing (Part III).
tfpt_3_e8_audit_bootstrap	E_8 audit & bootstrap. The seven E_8 slices as an audit raster ($E_6 \times A_2$ reads the flavor matrix) (Part I); the old cascade $D=60-2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}=5$ and the “8” in c_3 as overdetermined fixed points, only π irreducible (Part III).
tfpt_4_frontier	Frontier. The honest status of η_B , m_p/m_e , Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is <i>not</i> forced onto the ladder.
tfpt_research_contracts	Research contracts. The open gates as numbered contracts with lemma chains and closing theorems: (U_{wall}) (now reduced to the selector triangle: $(d, n) \Rightarrow R$ columnwise, residue = three atom pairings) and $G_{\text{metric}}/G_{\text{net}}$ (the index-4 seam-net identification — one theorem, three doors).
tfpt_horizon_readouts (App. H)	Appendix H — horizon unit system (reframe, not new physics). One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ($1920= W(D_5) $, $ \mu_4 =4$); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138).
origin_theory	Origin Theory. Why no free fundamental number remains: the $(g_{\text{car}}, N_{\text{fam}})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; [I]/[L] core plus one honestly-typed [P] cyclic interpretation.

The five formerly “research-level” problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\text{Spin}(10) \times \text{SU}(4)) / \Delta \mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a U_{point} anchor; θ_{12} the seam-misalignment residual). Seven companion files (plus this introduction), each result where it logically lives.

2 The architecture at a glance

The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.

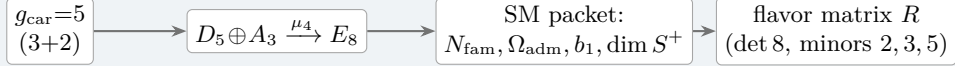
2 axioms (only π irreducible)



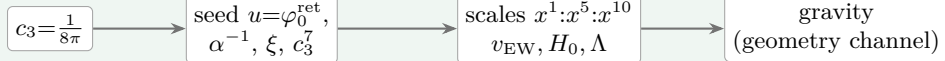
Key statement (the loop closes): previously D_5 , A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — and the red back-channel is the new part: the E_8 closure feeds back and *fixes the inputs*. $g_{\text{car}} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output form an overdetermined fixed point** — the structure does not “prove itself”; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.

Engine 1 — discrete closure (from $g_{\text{car}} = 5$)

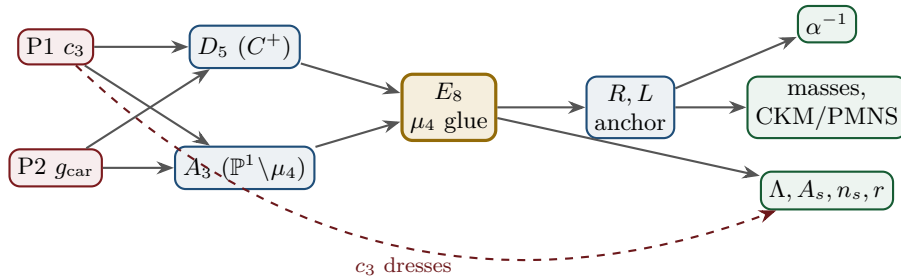


Engine 2 — boundary dressing (from $c_3 = \frac{1}{8\pi}$)



3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, $c_3=1/(8\pi)$	— (axiom)	[A]	c_3	no reflection-positive seam
P2	5-slot carrier $g_{\text{car}}=5$	— (axiom)	[A]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[I]	$\dim S^+=16, Y$	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[I]	$N_{\text{fam}}=3, \mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: disc $=\mathbb{Z}_4$, q -sum=2	D_5, A_3	[I]	240, 248	not even-unimodular
R, L	residue + winding matrix, anchor (1, 1, 2)	A_3	[I]	det 8, minors (2, 3, 5)	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha)=0$	$c_3, M=41$	[N]	137.0359992	no/second root (excl.)
masses	φ_0^{ret} -ladder $\lambda_Y^L \Lambda$	R, L, c_3	[I]/[P]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam $\varepsilon=c_3$	R, L	[N]/[P]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[P]	Λ, A_s, n_s, r	ambient RP fails

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early “flavor gate open” rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

The compact status formula (three honest layers)

compiler	admissible IR physics	strict physical TOE
closed	conditionally closed (RP, gap)	open

TFPT 5.0 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R + R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R -bridge amplitude normalisation U_{point} reduces to v_{geo} (Gate 1 closed, v75 — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-boundary measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (v77), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open). (i) *Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6 \log \frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([I]/[L] for the attractor, [verification/v56_unique_attractor.py]; the *physical identification* of the transport operator stays [P]), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — [verification/v6_bootstrap.py], [verification/v14_carrier_uniqueness.py]). (ii) *Newton’s constant is not an independent input:* fixing the physical Λ branch ($(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ relative to Λ , so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([I]/[P], [verification/v60_lambda_metrology_branch.py]), not predicted from pure numbers. What stays irreducible is then just π , the discrete E_8 -closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \rightarrow v_{\text{geo}}$, v75) and gravity ($1/G$, v68): the two [A] anchors are *one* (dimensional analysis; no metre from pure mathematics).

The hard reduction: everything open is two research gates plus typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

— one parabolic wall-selection, one quantum-gravity measure, and a set of deliberately typed QFT/cosmology interfaces. *Sharper aliases* (current state of the reductions; the historical gate names are kept for ledger continuity): $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, v49/v71/v75), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \rightarrow G_{\text{net}}$ — reduced to a single boundary-net theorem, “*the seam-Calderón net is the index-4 μ_4 simple-current extension of the carrier net*” (holomorphy then follows, v89/GATE.METRIC.06); since the QBL chain (v108–v113) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (v110), the certified channel counts the code identically (v112), pair transport is minimally complete (v111), and the one quasi-free kernel — rank = c : 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (v113), so closing G_{net} also closes the carrier selection [L]; $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers. *Latest reductions (v134–v140)*: the flavor selector residue R4' is now “derive *three atom pairings*” — $d = \frac{3}{2}a - 2 \cdot 1$ is pure anchor data and (d, n) forces R *columnwise* (v136/v139); the sheet/parity question GATE.QGEO collapses to *one $\mathbb{Z}_3$ choice* (the canonical map exists and is Schur-rigid, v137/v140); and the seam quantum clock R1 is *typed closed* — the reduced clock is an edge-sector object whose budget matches the external Volkov–Wipf edge exactly $(-\frac{8}{3} = 2 \times (-\frac{4}{3}))$, while the full 4d determinant is the gravity firewall (v138). The status card:

Item	Status	Reading
parameter-freeness (self-consistency)	[I]/[L]	gapped transport $\Rightarrow$ <i>unique</i> Perron–Frobenius attractor (v56); bootstrap web $g_{\text{car}}=5$ forced (v6/v14)
Newton constant G_N	[I]/[P]	not <i>independent</i> : branch pins $\bar{M}_{\text{Pl}}$ rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[I]	$g_{\text{car}} \ \text{Pl}_{1,a}(K) \ _1 / (N_{\text{fam}}^2 \Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
(U) flavor gate	[A] $\rightarrow v_{\text{geo}}$	Gate 1 closed : $U_{\text{point}} \rightarrow v_{\text{geo}}$ (ratios + Grand Mass Volume, v75); same anchor as $1/G$
$R + R^2$ gravity	[I]/[P]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
physical QG sector	[F]/[P]	gap-decoupled admissible measure ($\Delta_{\text{eff}}=1.648>0$)
ambient QG measure	[A](reduced)	Gate 2 tier B : holographically reduced to a finite seam- <i>boundary</i> measure (v76); IR tier closed (decoupling); strict-TOE cert. open (G_{metric})
Koide	[P]	$Q_{\text{src}} + \varphi_0/24$ as source→pole transfer conjecture
axion DM	[P]/[A]	$f_a = M_{\text{scal}}/128$ as a scenario
η_B	[P]	readout closed; leptogenesis Boltzmann solve missing
m_p/m_e	[A]	cross-sector QCD/EW ratio
N_*	[P]	reheating input, $N_* \simeq 57$ (continuous)
$a = (1, 1, 2)$	[L]/[P]	forced by $4 = 2 + 1 + 1$, conditional on the carrier
carrier selection $g=5$ (QBL)	[I]/[L] + [P]	theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = c); residue = the G_{net} premise itself — gate closes $\Rightarrow$ carrier [L]
selector establishment R4'	[I] + [P]	$(d, n) \Rightarrow R$ columnwise (v136); d pure anchor data, n unique from pairings (2, 8, 121) on the frame $(1, a, \sigma)$, $\det=11$ (v139); residue = derive the three pairings
sheet/parity GATE.QGEO	[I] + [P]	H^1 characters = $\text{Spec}(Q_+) = A_3$ exponents (v137); canonical map Schur-rigid; residue = <i>one $\mathbb{Z}_3$ choice</i> (v140)
seam quantum clock R1	[I] + [P]	typed closed: resummed clock = entropy power law (v124–v133); external VW edge = $2 \times$ reduced budget; full 4d det = firewall (v138)
E_8 Lean	[F] $\rightarrow$ [L]	hardening, not a blocker (script-certified)

Marker key: **[I]** exact identity · **[L]** Lie/lattice theorem · **[F]** formalised · **[N]** numerical fixed point · **[P]** physical/conditional · **[A]** axiom / open.

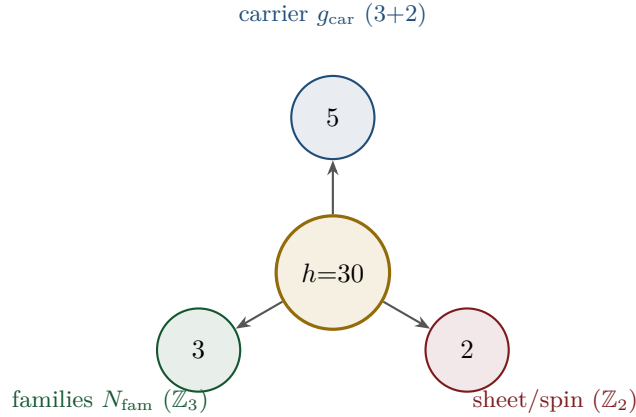
Only *two* research gates remain — (U_{wall}) and G_{metric} ; the frontier items are typed interfaces, not structural holes. Of these, (U_{wall}) is the only genuine *physics* gate left: G_{metric} is now heat-kernel grounded (G2: the spectral action gives $R + R^2$ structurally) and *gap-decoupled* (G5: $\Delta_{\text{eff}}=1.648>0$ isolates the closed admissible sector from the un-built ambient), so its residual — the ambient projective limit (G6) — is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority (U_{wall}) first. `[verification/v36_spectral_action_g2.py]`

Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ($N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) is a *consequence*. And even those *two tighten*: the bootstrap note shows $g_{\text{car}} = 5$ and the integer 8 in c_3 are *overdetermined E_8 -closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.)

4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

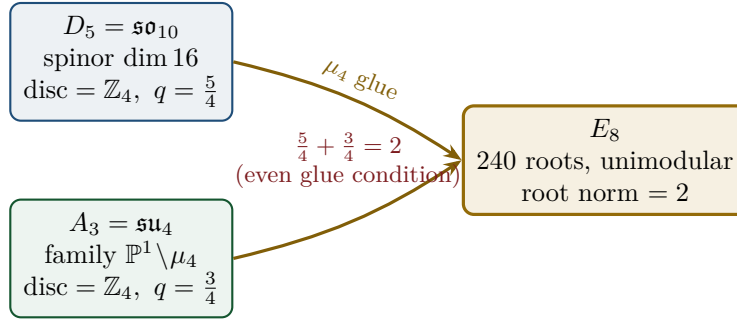


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. **[I]**
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. **[I]**
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30} ; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \setminus \mu_4$ is the A_3 Coxeter element. **[I]**

5 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$ is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [L]

6 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\text{car}}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier $3+2$, $\dim S^+ = 16$, $N_{\text{fam}} = 3$, $\Omega_{\text{adm}} = 48$	[I]
hypercharge	roots of the charge polynomial $bsY^2 + (s-b)Y - 1$	[I]
$U(1)$ index b_1	$b_1 = \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}$	[I]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha) = 0$	[I]/[N]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[I]/[P]
electroweak scale	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[I]
cosmological constant	$\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[I]/[P]
Hubble scale	$H_0 \sim \sqrt{\Lambda}$	[P]
inflation amplitude	$A_s = \frac{N_*^2}{24\pi^2} c_3^7$ in the R^2 branch	[P]
E_8	$D_5 \oplus A_3 + \mu_4\text{-glue}$	[L]

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1} \approx 137$ generates the electroweak scale (divided by $5 =$ carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

Typing note on Λ (a deliberate danger zone). “ Λ ” denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [I] readout), (ii) the dimensionless density quotient $\rho_\Lambda / \bar{M}_{\text{Pl}}^4$ (which carries a prefactor *branch*, hence [P]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier — not a separate module

With $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5 :

$$\boxed{\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \quad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = x^{10} R_\Lambda}, \quad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}.$$

EW is the unit x^1 , Hubble is the product over the $\binom{5}{1} = 5$ vertices, Λ is the product over the $\binom{5}{2} = 10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. “Hubble = slot product, Λ = pair product.” [1]

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \quad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}, \quad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers ($\text{tr} = 9 = N_{\text{fam}}^2$, $\det = 8 = h(D_5)$, principal 2-minors 2,3,5 with product $30 = h(E_8)$). Through the φ_0^{ret} -ladder $\hat{m} = \frac{v_{\text{geo}}}{\sqrt{2}} c(\varphi_0^{\text{ret}})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v49/v71), with only the *absolute* amplitude scale reducing to the (U_{wall}) anchor. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$; see below). (*Full derivation: document 2, tfpt_2_standard_model.*)

7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*, α^{-1} is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the C^+ spinor and the puncture geometry $\mathbb{P}^1 \setminus \mu_4$ produce D_5 and A_3 , glued to E_8 ; *topologically*, the common discriminant $\mathbb{Z}_4$ (the μ_4 winding) forces the glue and Φ_{30} /Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence.

Why the compiled architecture is plausible

It is (i) *parsimonious* (few independent assumptions $\Rightarrow$ low overfitting risk), (ii) *rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

8 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms $5/4$ and $3/4$ (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain “everything”; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8 .

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit:

$E_6 \times A_2$ reads the flavor matrix

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

9 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic $\leftrightarrow$ unitary representation); the algebraic selector R , the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\text{Spec}(Q_+)$ from D_4 , v69/v70; since v136/v139 the dual pair (d, n) forces R *columnwise*, d is pure anchor data, and $\text{Spec}(Q_+) = (1, 2, 3)$ is the A_3 -exponent grading on the seam curve’s cohomology, v137), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point} , an anchor; Research Contract 1). [\[I\]](#)/[\[A\]](#)
2. **Gravity — exact in the R^2 regime.** Spectral action $\rightarrow R+R^2$, BRST/Hodge kills the gauge directions, FRW reduction $\rightarrow$ Starobinsky; the only remainder is the *controlled* low-curvature truncation ($R^{n \geq 3} \rightarrow R^2$, valid for curvature $\ll M^2$). [\[I\]](#) in the regime
3. **Strong CP — decoupled.** $\theta_{\text{eff}}=0$ follows from three structural facts (γ_5 -Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [\[I\]](#) (given RP)
4. **Full dynamics — the admissible transfer model is closed.** The mass gap is the *explicit* discrete gap $6 \log \frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector; the continuum and metric-coupled theory still require the stated RP, cluster and limit assumptions. [\[I\]](#) (transfer model) / [\[P\]](#) (continuum)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (1886 jobs, 0 sorry, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [\[I\]](#)/[\[A\]](#)

In other words: the theory has shrunk from “many open audit points” to “*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms”. The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. (*Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.*)

The closure ledger — what is done, and what genuinely remains

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	Status	If not complete: what closes it
gauge group, N_{fam} , hypercharge, b_1	[I]	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	[L]	— complete
bootstrap $(g, \mu) = (5, 4)$, “8” in c_3	[L]	— complete (reverse glue $\mu^2 - 5\mu + 4 = 0$)
fine structure α^{-1}	[N]	— existence+uniqueness proved ; value is a numerical fixed point (ledger EM.FP.01), Ward origin [P]
fermion masses (φ_0^{ret} -ladder)	[I]/[A]	leptons exact; quark <i>ratios</i> closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$	[N]/[P]	θ_{12} cond. derived; θ_{23} maximal
flavour matrix / H2	[I]	dictionary <i>exact</i> (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); R forced columnwise by (d, n) (v136/v139); residue = three atom pairings
seam quantum clock (R1)	[I]/[P]	typed closed: reduced edge-sector clock (v124–v133); external VW edge = $2 \times$ reduced budget, full 4d det = firewall (v138)
inflation A_s, n_s, r , scalaron M	[N]	parameter-free given $M_{\text{Star}} = M_{\text{scal}}$
<i>genuinely open — the actual TOE remainder</i>		
dim. EW/QCD masses	[P]	standard RG threshold matching (no new physics)
QFT measure (admissible sector)	[I]	closed: RP from cusp spectrum, OS reconstruction
ambient QFT + full gravity (metric sector)	[A]	<i>not constructed here</i> (frontier doc); conditionally reduces to one named hypothesis (ambient RP) on the relative spectral-action measure
analytic boundary origin of π ($c_3 = \frac{1}{8\pi}$)	[P]	hardened: $c_3 = 1/(\mathbb{Z}_2 \oint_{S^2} K) = 1/(8\pi)$ (Gauss–Bonnet); “8” = rank E_8 ; π stays the one primitive
$\eta_B, m_p/m_e$, Koide, dark matter	[A]	frontier handles only (see document 4)

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret} -ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed); the *ambient/metric-sector* measure — i.e. *full quantum gravity* — is *not* constructed here and stays [A] (frontier doc), reducing conditionally to a single named hypothesis (ambient reflection positivity). The seam seed is *hardened*: $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete “8” equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and π is the lone continuous primitive.

10 Does this make the theory more probable?

Plausibility balance

Pro: (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors $\Rightarrow$ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit,

the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, modulo the seam-misalignment lemma). A final verdict “it maps reality” requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

11 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	TFPT value & derivation	Experiment (running / upcoming)	St.
solar angle $\sin^2 \theta_{12}$	$\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.3067$; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} \approx c_3$ (cond.)	JUNO (taking data since Aug 2025; sub-percent θ_{12}) — the sharpest live test	[N]/[P]
reactor angle $\sin^2 \theta_{13}$	$\varphi_0^{\text{ret}} e^{-5/6} = 0.0231$; seed $\times$ carrier-trace $e^{-\gamma}$, $\gamma = \frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[I]
atmospheric $\sin^2 \theta_{23}$	$\approx \frac{1}{2}$ ($\mu\tau$ -symmetric limit; octant not selected)	NOvA / T2K / DUNE (octant ambiguity, NuFIT 6.0)	[P]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)	CODATA-2022 137.035999177(21): dev. 2.9×10^{-10} from the recommended adjustment (1.9 σ of its uncertainty)	[I]/[N]
tensor ratio r	$r = \frac{12}{N_*^2} = 0.0040$ ($N_*=55$); R^2 attractor, $M=c_3^{7/2}\bar{M}_{\text{Pl}}$	now $r < 0.036$ (BICEP/Keck BK18); CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, obs. ~ 2033)	[P]
tilt n_s	$n_s = 1 - \frac{2}{N_*} = 0.964$ (same attractor)	Planck (0.965); CMB-S4 sharpens	[P]
amplitude A_s	$\frac{N_*^2}{24\pi^2} c_3^7 = 2.0 \times 10^{-9}$, parameter-free	Planck (2.1×10^{-9})	[P]
neutron EDM	$\theta_{\text{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)	PSI nEDM / SNS (running)	[I]
CP phase δ_{CKM}	$\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$ (frozen of record, v88); survives at $+0.98\sigma$ vs γ_{PDG}	LHCb / Belle II γ combination; decision at $\sigma_\gamma \leq 0.96^\circ$	[N]
cosmic birefringence β	$\beta = \frac{\varphi_0^{\text{ret}}}{4\pi} = 0.2424^\circ$ (seam readout)	ACT DR6 (0.4 σ agreement); Simons Observatory / LiteBIRD sharpen	[P]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	LEGEND / nEXO , DUNE, KATRIN	[P]
flavor invariants	$\det R = h(D_5) = 8$, principal minors (2, 3, 5), $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)	any future CKM/PMNS global fit must satisfy these	[I]
H_0 vs. Λ	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$, $\Lambda \sim e^{-2\alpha^{-1}}$, $H_0 \sim \sqrt{\Lambda}$: one engine	SH0ES / DESI / Planck (Hubble tension)	[P]

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

The two decisive near-term tests

(1) **JUNO and θ_{12} .** JUNO is *taking data since August 2025* and targets sub-percent $\sin^2 \theta_{12}$. TFPT commits to the conditional $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.3067$ (seam $\varepsilon = c_3$). A measured central value away from ~ 0.307 at high significance falsifies the seam mechanism — a *live* test.

(2) r . The R^2 scalaron with M fixed by c_3^7 predicts $r = 0.0040$ ($N_\star=55$), already below the current bound $r < 0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, first observations ~ 2033 –34). A detection of $r \gtrsim 0.01$ would be incompatible with the R^2 branch on which M_{Pl} and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate E_8 ; and the discipline rule “*every load-bearing number appears in at least one E_8 projection*” (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge itself is additionally *quantified* by an explicit null model: exhaustively enumerating a declared formula grammar (provably containing every scored TFPT formula) against conservative data windows gives a joint look-elsewhere-corrected null probability $\leq 10^{-30.7}$ that a random theory of equal formula complexity reproduces the scorecard — a statistical statement conditional on the declared grammar, not a claim of certainty [N] [verification/v100_numerology_null_mc.py].

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying M_{Pl}, A_s ;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\text{eff}} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [verification/v84_frozen_registry.py] re-derives each decimal from the two axioms on every suite run (formula $\leftrightarrow$ value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor (U_{point}) of the same nature as the one dimensional scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+ ;
- a sharper geometry $\leftrightarrow$ gravity bridge once the Hodge/ R^2 branch is lifted from [P] to [I].

Conclusion

TFPT’s sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\text{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious*, *more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation,

the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked [I] (exact identity), [L] (Lie/lattice theorem) or [N] (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\text{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map.

Script	What it machine-checks
<code>v1_e8_glue.py</code>	glue theorem $E_8=(D_5\oplus A_3)+\mu_4$: $\text{disc}=\mathbb{Z}_4$, $q(D_5)+q(A_3)=\frac{5}{4}+\frac{3}{4}=2$, $240=16\cdot 5\cdot 3$, $248=240+8$
<code>v2_carrier_pascal.py</code>	$g_{\text{car}}=5$ unique Pascal solution; $16=1+5+10$, $N_{\text{fam}}=3$, $\text{rank } E_8=8$, $\Omega_{\text{adm}}=48$, $b_1=\frac{41}{10}$
<code>v3_em_alpha.py</code>	EM closure: unique root of $F_{U(1)}(\alpha)=0$, $\alpha^{-1}=137.0359992168$, existence/uniqueness, inverse test, ablation
<code>v4_flavor_matrix.py</code>	residue matrix R : $\det=8$, minors $\{2, 3, 5\}$, $\chi_R=t^3-9t^2+10t-8$, SNF $\text{diag}(1, 1, 8)$, $\ R\ _F^2=78$, $\sum L=40$
<code>v5_e8_cascade.py</code>	cascade $D_n=60-2n$: endpoints, exponent rungs $\rightarrow 240$, IR tail 46080, variance 6552
<code>v6_bootstrap.py</code>	reverse glue $\mu^2-5\mu+4=0$; $g_{\text{car}}=5$ three ways; $8=\text{rank } E_8=h(D_5)=\varphi(30)$
<code>v7_gravity_cosmo.py</code>	scalon exponent 7, $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$, A_s, n_s, r , Ω_b , $\sin^2 \theta_{12}$
<code>v8_horizon.py</code>	horizon code: $\frac{1}{2\pi}=4c_3$, $1920= W(D_5) $, S_{dS} , Page, $\beta_{\text{rad}}=0.2424^\circ$
<code>v9_neutrino_texture.py</code>	polar angle: explicit $\mu\tau$ Majorana texture $\rightarrow \sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$, $\theta_{23}=45^\circ$, $\theta_{13}=0$
<code>v10_projection_involutions.py</code>	Q, Σ algebra: $K=R+Q\Sigma$, $L=R+Q(I+\Sigma)$; χ_Q, χ_K ; det ladder $3\cdot 4\cdot 8\cdot 20=1920= W(D_5) $; $a^\top K a=41$
<code>v11_unique_KQ.py</code>	K, Q are the <i>unique</i> nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
<code>v12_mass_generation_polynomials.py</code>	sector/generation polynomials of K and the anchor-block det ladder $(9, 10, 16, 40)$
<code>v13_open_gates.py</code>	gate closures: $M=41=10b_1=\sum L+N_\Phi$; $Q_+=A_3$ exponents, $Q_-^2=N_{\text{fam}}$
<code>v14_carrier_uniqueness.py</code>	$g_{\text{car}}=5$ unique ($N_{\text{fam}}\in\mathbb{Z}^+$, $\text{rank } E_8=8$); split $(3, 2)$ from $b+s=5$, $b^2+s^2=13$; $\text{Tr } Y=\text{Tr } Y^3=0$
<code>v15_bootstrap_classification.py</code>	$D_5\oplus A_3$ unique familyful cyclic-glue of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
<code>v16_solar_dual_anchor.py</code>	$a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$; $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$; full PMNS
<code>v17_hexagonal_resolvent.py</code>	finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark c backbone)
<code>v18_quark_yukawa.py</code>	quark source ratios $\frac{m_u}{m_d}=0.470085$, $\frac{m_c}{m_s}=13.61$, $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$; full hierarchy
<code>v19_monodromy_moduli.py</code>	exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4 -fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
<code>v20_lepton_c_derivation.py</code>	lepton c 's derived: $\delta=\frac{1}{2}$ resolvent $c= \mu_4 ^w/(\frac{5}{4}-\cos\frac{r\pi}{3})\Rightarrow\frac{16}{7}, \frac{4}{3}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}=\frac{32}{9}\Rightarrow\frac{7}{6}$
<code>v21_solar_product_quark.py</code>	solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$), $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$; quark law fails ($\frac{1}{7}$ vs 0.724)

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Script	What it machine-checks (<i>continued</i>)
v22_open_gates_audit.py	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
v23_anchor_generator.py	anchor $a=(1,1,2)$: $e_k=(4,5,2)$, $c_3=\frac{1}{2e_1\pi}$; $p_n=2+2^n \Rightarrow 240, 248$, binary ladder; inputs $\rightarrow \{a, \pi\}$
v24_quark_ratio_closure.py	quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$; rational c gauge, Λ_q in $O(1)$
v25_frontier_conjectures.py	conjectures: Koide $Q_{\text{src}} + \frac{\varphi_0^{\text{ret}}}{24} \approx \frac{2}{3}$; axion $f_a = M_{\text{scal}}/128 \rightarrow m_a \approx 23.8 \mu\text{eV}$
v26_flavor_frontier_unification.py	the “11” is not uniquely forced ($\Rightarrow$ [P]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
v27_wall_representative.py	explicit balanced wall rep W_{wall} (cols = perm(0,1,2), rows $w=(2,1,1)$, pardeg 0); selector $\det R=8$, $\text{Spec}(Q_+)=\{1,2,3\}$
v28_gravity_fR.py	$R+R^2$ closed: $f(R)$, $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ scalaron, $N_\star=57 \rightarrow n_s, r, A_s$; full metric measure open; $248c_3^2 < 6 \log \frac{3}{2}$
v29_research_contract_certs.py	research-contract certificates: $C_U^{(1)}$ wall enumeration (1296 $\rightarrow$ 144 $\rightarrow$ 5 orbits; symmetry alone does not pick W_{wall}) and $G5$ gap-dominance $2\ V\ < \Delta$
v30_d4_character_variety.py	$C_U^{(2)}$: full D_4 -fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the $R4'$ selector establishment)
v31_R_dictionary.py	$R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); R integer but ρ -invariants continuous $\Rightarrow R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)
v32_rh_splitting.py	RH route: $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ exact $\Rightarrow$ splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \Leftrightarrow \text{diag}(A_0) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ (algebraic); $\prod M_k = I + R$ -bridge stay open (c_u/c_d not forced)
v33_explicit_flat_bundle.py	explicit valid (U_{wall}) flat bundle (RH solve): cusp + $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 + \ M_\infty - I\ \sim 10^{-9}$ ($\prod M_k = I$) + irreducible (case A)
v34_h2_bridge_attempt.py	H2 bridge: explicit per-puncture M_k (cusp, $\prod M_k = I$); honest negative — natural extraction $\neq$ lepton amplitudes, the Γ^{min} dictionary is the remaining input (c_u/c_d not obtained)
v35_quark_qcd_boundary.py	the open gates are the discrete $\leftrightarrow$ continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)
v36_spectral_action_g2.py	G2: spectral action $\rightarrow R+R^2$ ($a_2 \sim -R/3$, $a_4 _{R^2} \sim R^2/72$); $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$, c_3^7 fixes f_0 ; gap-decoupling $\Delta_{\text{eff}} = 1.648 > 0 \Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target
v37_plucker_anchor.py	anchor-plane Plücker apparatus: $\ Pl(K)\ _1=11$, $\ Pl_R(K)\ _1=26$, $\sum Pl_R(L)=60$; pencil $\det(K+xQ)=3x^3+7x^2+6x+4$; $K_e-K_u=a$, $L_e-L_u=\text{Exp}(A_3)$; lepton ring $c_e c_\tau = \mathbb{Z}_2 c_\mu$; Koide $u \rightarrow \frac{53}{54}u$ ([P])
v38_uwall_killswitch.py	(U_{wall}) U2 kill-switch hardened: D_4 -cusp rep irreducible at all 400 sampled points (commutant=1), mixing ≥ 0.35 , reducible locus measure-zero $\Rightarrow$ case A generic, (U_{wall}) survives; amplitude dictionary still research-level
v39_uwall_selectors.py	(U_{wall}) selector side closed on the explicit point: splitting type = anchor $a=(1,1,2)$, pardeg=0; $\det R=8=n \cdot a$, $\text{Spec}(Q_+)=\{1,2,3\}=3\alpha+1$ read off the bundle; only the amplitudes c_u/c_d (Γ^{min} Hitchin) remain
v40_harmonic_metric.py	(U_{wall}) harmonic metric = FINITE linear algebra (polystable $\Rightarrow$ unitary $\Rightarrow\Phi=0$): unique invariant form $H>0$, unitarised $\widetilde{M}_k$ cusp-class with $ \widetilde{M}_k \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$; the feared Hitchin PDE collapses to unitarisation + combinatorial R

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Script	What it machine-checks (<i>continued</i>)
v41_leg_assignment.py	(U_{wall}) final leg test: amplitudes = $\delta = \frac{1}{2}$ resolvent (closed); $\Lambda_\mu > 1$ rules out holonomy moduli; harmonic holonomy $ \text{diag} = (0, \frac{1}{2}, \frac{1}{2})$ supplies $\delta = \frac{1}{2}$ (suggestive); honest negative on literal c_u/c_d reproduction
v42_exterior_leg.py	(U_{wall}) Exterior Leg Lemma: u, d share $(r, w) = (1, 1)$ so any scalar leg gives $c_u/c_d \equiv 1$; the u/d split is the $\Lambda^2 F$ area $\text{Pl}_{1,a}(K)$, $\ \cdot\ _1 = 11$ generated $\Rightarrow \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117}$ [I]; phys. identification [P]
v43_exterior_bridge.py	(U_{wall}) $\Lambda^2 F$ bridge: $\Lambda^2(\widetilde{M}) = \widetilde{M}$ (exterior leg = $\bar{3}$) confirmed; continuous action $\neq$ integer $\text{Pl}(K) \Rightarrow$ bridge is the discrete H2 invariant ($\det R = 8$, $\text{Spec}(Q_+)$ confirmed), not continuous
v44_carrier_exterior.py	(U_{wall}) Lie-level grounding: $16 = \Lambda^{\text{even}}(5) = 1 + 10 + 5$; $u^c \in \Lambda^2(5)$, $d^c \in \Lambda^4(5) \Rightarrow$ exterior degrees $(2, 4)$, $\text{sum} = 6 = R^+(A_3) $, $\text{diff} = 2 = \mathbb{Z}_2 $; the exterior leg is the carrier's own Λ^2 grading (no special math)
v45_family_exterior.py	(U_{wall}) the “11” is Pascal: $ \text{Pl}(K) = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(\mu_4)$; $\ \text{Pl}(K)\ _1 = 11 = \sum_{k \leq 2} \binom{4}{k} = 16 - g_{\text{car}}$ (cumulative ext of μ_4 up to $\deg u^c = 2$); $15 = \dim \mathfrak{su}(4) = 10 + 5 = N_{\text{fam}} g_{\text{car}}$
v46_grand_mass_volume.py	absolute mass-scaling is [I], not a product rule: sector $\det(\varphi_0^{\text{ret}})^6, (\varphi_0^{\text{ret}})^9, (\varphi_0^{\text{ret}})^{10}$ (exponents $ R^+ A_3 , N_{\text{fam}}^2, \Lambda_\Lambda \Rightarrow \det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{25} = (\varphi_0^{\text{ret}})^{g_{\text{car}}}$; Q -rows $(4, 5, 6)$ $\text{sum } 15 = \dim A_3$; $25 + 15 = 40 = \sum L$
v47_selection_theorem.py	Boundary Carrier Selection (Thm A): $\mu_4 \rightarrow A_3$, $N_{\text{fam}} = 3$; 16-spinor $\rightarrow D_5$; $q(D_5) + q(A_3) = 2$, glue $4 = \mu_4 $; only $D_5 \oplus A_3$ meets all four conditions ($D_8, E_7 + A_1, E_6 + A_2, A_4 + A_4, A_8$ excluded) [L]/[P]
v48_em_ward.py	EM boundary Ward (Thm C): $F_{U(1)} = \text{Maxwell } \alpha^3 - \text{Calderón } 2c_3^2 \alpha^2 - \text{transport } 8b_1 c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$; $8b_1 = \frac{4}{5} M = \frac{164}{5}$, $-\frac{5}{4} = q(D_5)$ [I]; Ward origin [P]
v49_readout_rigidity.py	Readout Rigidity (Thm U2): $c_u/c_d = \frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q} \Rightarrow$ no point-uniqueness needed; ($U_{\text{wall}} = U_{\text{unitary}} + U_{\text{H2}} + U_{\Lambda^2} + U_{\text{point}}$, only U_{point} open
v50_q_geometry.py	Q geometry (Thm Q): $Q_+ = \text{diag}(3, 2, 1)$ -block $\chi = (t-1)(t-2)(t-3)$, $Q_- \chi = t(t^2-3)$; Q unique under rows $(4, 5, 6)$ /cols $(9, 5, 1)$ /spectra; geom. origin [P]
v51_boundary_half_step.py	glue norms $= (g_{\text{car}}, N_{\text{fam}})/ \mu_4 $: $q(D_5) = \frac{5}{4}$, $q(A_3) = \frac{3}{4}$; four ops forced $\rightarrow \{\text{sum} = 2 \text{ root norm, diff} = \frac{1}{2} = \mathbb{Z}_2 / \mu_4 = \delta, \text{prod} = \frac{15}{16}, \text{ratio} = \frac{5}{3}\}$. So $\delta = \frac{1}{2}$ is the carrier-family count over the glue index, not chosen [I]
v52_pencil_endpoints.py	$K + xQ$ pencil endpoints: $P(-1) = 2 = \mathbb{Z}_2 $, $P(0) = 4 = \mu_4 $, $P(1) = 20 = \det L$, $P(2) = 68 = 2p_5(a)$; $\det(K - Q) = 2$, $\text{tr} = N_{\text{fam}}$ (audit ext. of v37) [I]
v53_compiler_core.py	Big picture: the whole integer skeleton from $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ — rank $E_8 = 8$, $ \mathbb{Z}_2 = 2$, Pythagorean $\Delta_Y = g_{\text{car}}^2 = 25 = N_{\text{fam}}^2 + \mathbb{Z}_2 \cdot \text{rank } E_8 = 9 + 16 = \dim S^+$ (diff. of squares); anchor char-poly $(t-1)^2(t-2)$ unique [I]
v54_seam_horizon_keystones.py	The “8” in $c_3 = \frac{1}{8\pi}$ overdetermined and gravitationally aligned ($2 \mu_4 = \text{rank } E_8 = h(D_5) = \varphi(30) = 8\pi$ Hawking/Einstein); one boundary transport operator ($\text{spec } \{1, (2/3)^6, (1/3)^6\}$, $\lambda_2 = (2/3)^6$) governs both SM flavor gap and horizon Page recovery [I]
v55_coxeter_cycle.py	Computed E_8 Coxeter element: order exactly $30 = \mathbb{Z}_2 N_{\text{fam}} g_{\text{car}}$, eigenphases = exponents = totatives of 30, so $\text{rank } E_8 = 8 = \varphi(30) = \text{live phases of the order-30 cycle}$; one α^{-1} sets EM, S_{dS} and Λ ($S_{dS} \rho_\Lambda = 32\pi^4$) [I]/[L]
v56_unique_attractor.py	Gapped boundary transport (gap $6 \log \frac{3}{2} > 0$) $\Rightarrow$ unique attracting fixed point, geometric rate $(2/3)^6$ (two starts $\rightarrow$ same direction); Coxeter in $ \mu_4 = 4$ planes; entropy reset = rank-1 projector [I]/[L]
v57_horizon_crosslinks.py	Horizon cross-links: $c_3 = \frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ($1/4 = 1/ \mu_4 $ entropy density); Hod QNM $\omega_R/T_H = \ln 3 = \ln N_{\text{fam}}$; Hawking power denom $1920 = W(D_5) $; one α^{-1} for EM/ S_{dS} / Λ /Hubble [I] (+ [P] identification)

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Script	What it machine-checks (<i>continued</i>)
v58_seam_horizon_chain.py	Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided S^2 Gauss–Bonnet $c_3=1/(2\cdot 4\pi)=\frac{1}{8\pi}$; KMS $4c_3\kappa$; $S_{BH}=M^2/2c_3$; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) [A] open
v59_area_law_evidence.py	Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim \frac{\epsilon}{3} \log L$), by the <i>same</i> gap that fixes the attractor; $8= \mathbb{Z}_2 \mu_4 $. The c_3 coefficient stays [A] open
v60_lambda_metrology_branch.py	Λ branch selection: $(8\pi)^2\delta_{\text{top}}=\frac{3}{4\pi^2}$, mis-scale $2c_3/\delta_{\text{top}}=\frac{64\pi^3}{3}=661.47 \Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215^\circ \pm 0.074^\circ$ vs 0.2424° (0.37σ) [I] / [N]
v61_cft_bridge.py	Boundary-CFT mirror: level-1 WZW central charges $c(E_8)=8$, $c(D_5)=5$, $c(A_3)=3$ = (rank $E_8, g_{\text{car}}, N_{\text{fam}}$); conformal embedding $c_{\text{coset}}=0$; $SU(3) \leftrightarrow SU(4)$ reconciliation $N_{\text{fam}}= \mu_4 -1$, $4 \rightarrow 3+1$ (one geometry $\mathbb{P}^1 \setminus \mu_4$, two sides) [I] / [L]
v62_data_scorecard.py	TFPT vs 2024/25 data: α^{-1} , $\sin^2 \theta_{12}$, β_{rad} match; $\sin^2 \theta_{13}$ (2.0σ) and Starobinsky n_s vs DESI ($\sim 2\sigma$) mild tensions; θ_{23} open; $r \approx 0.004$ future falsifier [N]
v63_seam_engineering_index.py	Exact Seam-Engineering Index $\Xi=2\ V\ /\Delta=\frac{31}{24\pi^2 \log(3/2)} \approx 0.323$ ($2\ V\ =\frac{31}{4\pi^2}$, $248=\dim E_8$, $31=1+h^\vee$), $\Delta_{\text{eff}} \approx 1.648$; four-class possibility taxonomy [I] / [A]
v64_causal_boundary_nogo.py	Causal Boundary Engineering <i>conditional no-go</i> : gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives [I] (core)/ [P] (chain)
v65_falsification_layer.py	Prediction layer with explicit failure thresholds: 3 core ($v_{\text{GW}}=c$, no 4th gen, seam=8), 4 observational ($\beta, r, n_s, \theta_{12}$), no-go + unitarity; $N_\star$ demoted to [P] input; none violated [N]
v66_e8_casimir_degrees.py	Organizing simplification: the compiler atoms <i>are</i> the E_8 Casimir degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ ($2= \mathbb{Z}_2 $, $8=\text{rank}$, $14=\dim G_2$, $20=\det L$, $24= W(A_3) $, $30=h(E_8)$); $\sum=128=2^7$ (dS prefactor), $\sum \exp=120= R^+(E_8) $ [L]
v67_area_law_coefficient.py	Central-theorem <i>structural closure</i> : Fursaev–Solodukhin $S=4\pi kA$, $k=c_3/2 \Rightarrow S=2\pi c_3 A=\frac{1}{4}A$ ($2\pi c_3=\frac{1}{4}$); $c_3=\frac{1}{8\pi}$ is the <i>unique</i> value with replica-coeff = BH $\frac{1}{4}$. Reduces to the one coeff $k=c_3/2$ (then <i>forced</i> , v73) [I] / [L]
v68_seeley_dewitt_residual.py	Residual <i>resolved as an anchor</i> : $a_2=-\frac{d}{192\pi^2}R \Rightarrow$ the <i>absolute</i> $\frac{1}{16\pi G}$ is UV-sensitive ($\propto d\Lambda^2 f_2$), so the <i>scale</i> $1/G$ is the one dimensionful anchor (the dimensionless $k=c_3/2$ is forced, v73); cutoff-indep. $R^2/\text{scalaron}$ derived [I] / [L]
v69_d4_q_geometry.py	D_4 -equivariant Q-geometry (gate [P] $\rightarrow$ [L]): on $H_1(\mathbb{P}^1 \setminus \mu_4)=B_1 \oplus E$, $Q_+=3w+1=\text{diag}(1, 2, 3)$ (cusp weights on $\mu_4=\mathbb{Z}_4$ eigenspaces), $Q_-=E$ -coupling $\sqrt{N_{\text{fam}}} \Rightarrow t(t^2-3)$; Σ -split $=D_4=\mathbb{Z}_4 \rtimes \mathbb{Z}_2$ [L]
v70_q_integer_lift.py	Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action R unimodular (eigenvalues $\{-1, i, -i\}$); $\det Q=3=N_{\text{fam}}$, SNF $\text{diag}(1, 1, 3) \Rightarrow$ residual = one lattice invariant $\det Q=N_{\text{fam}}$, not closable by D_4 alone [I] / [L]
v71_simple_r_bridge.py	<i>Simple</i> R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ($\det R=8=\text{rank } E_8$ lattice + $\text{Spec}(Q_+)=\{1, 2, 3\}$ from v69) $\Rightarrow$ quark <i>ratios</i> are pure integer Plücker readouts ($\frac{55}{117}$, no monodromy, v49 rigidity); residual U_{point} = absolute norm. = an anchor [I] / [A]
v72_q_det_from_cusp.py	$\det Q=N_{\text{fam}}$ derived from the cusp class: $\det Q= \text{coker } Q =\text{cusp-class order}=\text{denominator of the weights } \{0, \frac{1}{3}, \frac{2}{3}\}$ (same data as $\text{Spec } Q_+$); $\text{coker } Q=\mathbb{Z}/N_{\text{fam}}$ = triality deck group $\Rightarrow$ integer-lift residual closed [I] / [L]
v73_k_c3_half.py	Central coefficient $k=c_3/2$ <i>forced</i> : $(1/2)$ variational $\times 1/(\mathbb{Z}_2 2\pi\chi(S^2))$ Gauss–Bonnet topology ($\chi(S^2)=2$), both cutoff-indep $\Rightarrow$ Fursaev–Solodukhin $S=A/4$; only the absolute 4D Newton scale stays the anchor (v68) [I] / [L]

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Script	What it machine-checks (<i>continued</i>)
v74_compiler_micro_lemmas.py	Spine quotient ladder ($\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3}$ = adjacent quotients of (2, 3, 4, 5)); pencil differences (2, 16, 48) = ($ \mathbb{Z}_2 $, $\dim S^+$, Ω_{adm}) sheet $\rightarrow$ 1gen $\rightarrow$ 3gens; anchor QF $a^\top K a - \mathbf{1}^\top K \mathbf{1} = 41 - 25 = 16 = \dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L = R + 61e_1^\top$, $a^\top R^{-1} \mathbf{1} = 0$ (Sherman–Morrison rank-one invariance) [I]
v75_upoint_to_vgeo.py	Gate 1 complete: $U_{\text{point}} \rightarrow v_{\text{geo}}$. Ratios closed (v49/v71) + lepton c exact (v20) + sector products = φ_0^{ret} -powers (v46); (ratios, product) $\Leftrightarrow$ (individuals) bijection $\Rightarrow$ absolute amplitudes fixed up to <i>one</i> scale v_{geo} = the same anchor as $1/G$ (v68); two $[A] \rightarrow$ one [I]/[A]
v76_gmetric_reduction.py	Gate 2: Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, $\ V\ \leq 248c_3^2 = \frac{31}{8\pi^2}$); Tier B (strict TOE) holographically reduced from bulk to a finite seam- <i>boundary</i> measure (conditional: RP+tightness $\Rightarrow$ closes) [I]/[P]
v77_e8_conformal_net.py	G6 route: the seam boundary measure = the $(E_8)_1$ <i>lattice net</i> (rigorous, holomorphic $c=8$); level-1 $c(E_8) = \frac{248}{31} = 8$, $c(D_5)=5$, $c(A_3)=3$, embedding $5+3=8$ (coset $c=0$); imports G_6 into RCF/T/conformal-net rigor [I]/[P]
v78_vgeo_floor.py	v_{geo} floor: no pure number is dimensionful $\Rightarrow v_{\text{geo}}$ irreducible by logic; all scales are ratios $\times$ one M_{Pl} ; $S_{dS} \rho_\Lambda = 32\pi^4$ pins it from one measurement; theory at the minimum (one scale + π) [I]/[A]
v79_review_identities.py	Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n = (3^n - (-2)^n)/5 = 1, 1, 7, 13, 55, 133$ (roots = carrier charges); Inverse Anchor Thm ($\mathbf{1}^\top M^{-1} \mathbf{1} = 1/\text{atom}$, $a^\top M^{-1} a = 1$ for R, K, L); MacWilliams (1, 10, 5); $6Y^2 - Y - 1 = (2Y - 1)(3Y + 1)$. Rejects m_p/m_e , η_B near-formulas (firewall) [I]/[L]
v80_operator_pencil_geometry.py	Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ} = (N_{\text{fam}}x + \mathbb{Z}_2)(N_{\text{fam}}x + g_{\text{car}}) = (3x+2)(3x+5)$ ($\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain (3, 10, 16, 20); type checker $\det B_{Q,K,R,L} = (9, 10, 16, 40)$; audit: differential spectrum, $F_4 \times G_2$ shadow $248 = 52 + 14 + 26 \cdot 7$ [I]/[L] + [P]
v81_singular_pencil_matrices.py	Double cover $y^2 = \det B_{K+xQ} = (3x+2)(3x+5)$: Koide $x = -\frac{2}{3}$ and carrier $x = -\frac{5}{3}$ are the two branch points (deck $2 = \mathbb{Z}_2 $; disc $= 81 = N_{\text{fam}}^4$; separation $= 1$ transport period). Clearing matrices $C_{2/3} = 3K - 2Q$ ($D_5 \oplus A_3$ glue, $B = (5, 7)^\top (9, 11)$, $\sum = 240$), $C_{5/3} = 3K - 5Q$ (neutral, $\chi = (\lambda+2)(\lambda^2 + \lambda + 6)$); scalaron $7 =$ mixed anchor disc $/N_{\text{fam}}$ [I]/[L]
v82_koide_attractor_splitting.py	Koide attractor <i>forced:</i> the branch-divisor-preserving Möbius map fixing $q=2, 5$ is unique, multiplier $(2/3)^6 = \lambda_2$ of the established transfer operator $\Rightarrow F'(2) < 1$ attractor, $F'(5) > 1$ repeller; three postulates $\rightarrow$ one [P]. Splitting trichotomy disc $81/49/40$: the clean split is non-generic (anchor-first) [I] + [P]
v83_e8net_holomorphic_uniqueness.py	Holomorphy pins $(E_8)_1$: #primaries = det Cartan ($D_8:4$ vs $E_8:1$), and a holomorphic $c=8$ chiral CFT is the lattice theory of the <i>unique</i> even unimodular rank-8 lattice (Minkowski–Siegel mass $1/ W(E_8) $); carrier Pascal truncation $K = (g-1)/2 =$ half-spinor split [L]/[I]
v84_frozen_registry.py	Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula $\leftrightarrow$ value): exactly <i>one</i> θ_{12} prediction of record; r , n_s as $N_\star$ <i>bands</i> ; conditional entries carry their hypothesis by name [N]
v85_master_cover.py	Master-cover theorem: $\det B$ is $\text{GL}(2)$ -covariant on $\text{span}\{K, Q\} \Rightarrow$ exactly <i>one</i> double cover up to Möbius (disc $= N_{\text{fam}}^4 \det(G)^2$); the disc-81 family is its orbit ($-\frac{8}{3} = -\text{rank } E_8 / N_{\text{fam}} =$ carrier – one period); μ_4 is <i>not</i> a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron <i>scale</i> [I]/[L]
v86_nstar_reheating.py	$N_\star$ band $\rightarrow$ point [P]: $\Gamma = 4M^3/48\pi \bar{M}^2 = 128 \text{ GeV}$, $T_{\text{reh}} = 9.6 \times 10^9 \text{ GeV}$, $N_\star(0.05/\text{Mpc}) = 51.4 \Rightarrow n_s = 0.9611$, $r = 0.0045$ (inside the frozen band); honest tensions: $n_s + 0.9\sigma$, A_s dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [P]

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Script	What it machine-checks (<i>continued</i>)
v87_bulk_uniqueness_reduction.py	Red-team Target A residual (ii) conditionally closed: holomorphic $\Rightarrow$ Rep = Vect $\Rightarrow$ 2D bulk <i>unique</i> (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits <i>six</i> modular invariants (incl. both E_8 pairings) $\Rightarrow$ Target A = one residual [L] + [P] / [A]
v88_cp_phase_audit.py	Target-D CP residual quantified: frozen $\delta=\pi/3+3\lambda^2$ survives at $+0.98\sigma$ vs γ_{PDG} ; the 0.07° -near alternative $\pi/3+2\lambda^2$ recorded as look-elsewhere trap, <i>not</i> adopted; decision at $\sigma_\gamma \leq 0.96^\circ$ [N]
v89_carrier_index_lemma.py	Carrier Index Lemma: KLM $\mu_A=[B:A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1:(D_5)_1 \times (A_3)_1]=4= \mu_4 $ (glue order <i>is</i> the index); μ -additivity $16/4^2=1 \Rightarrow$ holomorphy <i>follows</i> ; Gate A (H) $\Leftrightarrow$ (H') index-4 extension [L]
v90_conical_defect_chain.py	Fursaev–Solodukhin factor 4π <i>derived</i> (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S=4\pi kA$, $k=c_3/2 \Rightarrow S=A/4 \Leftrightarrow c_3=\frac{1}{8\pi}$ (sympy-solved); residual isolated to “seam determinant $\Rightarrow$ EH form” [I] + [A]
v91_spine_tetrahedron.py	Spine tetrahedron $\{2,3,4,5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120= R^+(E_8) $; $240= \mu_4 E(K_4) E(K_5) $ with K_6 negative control; dual cuts typed tautological; sub-grammar incompleteness honest [I]
v92_glue_uniqueness.py	Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two <i>chiralities</i> , swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$; tower carrier $\rightarrow SO(16)_1 \rightarrow (E_8)_1$, nothing else [I] / [L]
v93_koide_relaxation_toy.py	Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$; source at $\rho=-\varphi_0^{\text{ret}}/24$ to 0.8% ([P]); honest negatives: flow time $t=2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a <i>continuous</i> generator [I] / [N] + [P]
v94_sheet_diamond.py	Sheet diamond: all four flavor operators on <i>one</i> surface $M(s,t)=R+Q \text{diag}(s,t,t)$ (pencil = the cut $(1+x, x-1)$; diagonal-cut disc 153 negative control); winding line $\det B=2 \det$ for all s ; cofactor seam normal $(5, -9, 6)$; Plücker canonicalisation: 11 and 26 both live in K ; spine-quotient firewall <i>rejected</i> [I]
v95_centered_diamond.py	Centered cross: $Q=U+V$, $R/L=C \mp U$, $K/F=C \mp V$ (five objects $\rightarrow$ three); Spec $V=\{0,1,2\}=N_{\text{fam}} \cdot \text{cusp}$; $\det C=14$, $\sum C=31=2^{g_{\text{car}}}-1$; $\text{Pl}_R(C)=7(2,3,1) \parallel \text{Pl}_R(L)=10(2,3,1)$; G_2 reading audit-typed [I]
v96_branch_kernel_selection.py	Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have <i>integer</i> kernels (carrier kernel = 1); collapse lemma $\Rightarrow (-1, 1, 0)$: up = -down, <i>lepton pairing</i> = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$); anchor-placement controls [I] / [L]
v97_sheet_conjugation_bridge.py	Sheet-conjugation bridge (P1 $\rightarrow$ one [P]): the anchor forces the cusp conjugation uniquely $\Rightarrow$ integer T_A , $\det=-1$; $a=e_2+e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\text{span}\{\mathbf{1}, a\}$; $\langle T_A, \Sigma \rangle = D_4$; sheet-index lemma (intertwiner dets even, $\min 2= \mathbb{Z}_2 $); remaining [P] : Q_+ grading = A_3 discriminant grading [I] / [L] + [P]
v98_discriminant_dictionary.py	Dictionary <i>derived</i> (P1 closed modulo GATE.QGEO): $G=T_A\Sigma$ acts integrally on the cusp basis ($Ge_1=-e_1$, $Ge_3=e_2$, $Ge_2=-e_3 = B_1 \oplus E$); cusp-0 line = self-conjugate $\mathbb{Z}_4$ -class-2 line; $T_A GT_A^{-1}=G^{-1}$ ($k \mapsto -k$); $q_{A_3}(1)=q_{A_3}(3)=\frac{3}{8}$, $q_{A_3}(2)=\frac{1}{2}$; reflection classes = glue-swap vs glue-fix parities $\Rightarrow$ P1 carries no separate [P] ; residual = the one existing Q -geometry gate [I] / [L]
v99_koide_flow_time.py	Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\text{fam}})(q-2)(q-5)=(\Delta/N_{\text{fam}})\det B(q)$, time-1 map = F [I] ; non-integrality of $t=2.84$ is <i>data-fragile</i> (Q crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$, t passes every integer ≥ 3 within $+0.5\sigma$) [N] ; conditional steps: $n=2$ excluded (-2.9σ), $n=3=N_{\text{fam}} \Rightarrow m_\tau=1776.9427 \text{ MeV}$ ($+0.14\sigma$), decision at $\sigma(m_\tau) \sim 0.01 \text{ MeV}$ [P] ; $\ln(46080)$ scale reading destroyed by controls (firewall)

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Script	What it machine-checks (<i>continued</i>)
v101_horizon_anchor.py	The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$, roots $(1, 1, -2)$ = traceless anchor; $S_{\text{Nariai}}/S_{\text{dS}}=\frac{2}{3}= \mathbb{Z}_2 /N_{\text{fam}}$ (per horizon $\frac{1}{3}$); interpolation $(x^2+1)/\Phi_3(x)$; three-sheet total $ \mathbb{Z}_2 S_{\text{dS}}$ conserved; $Q_{\text{geom}} \in [\frac{3}{8}, \frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$, deck = horizon swap; $\tau_{BH}=128 g_{\text{car}} M^3/c_3$; six atom landings, zero free parameters [I] + [P] (bulk reading)
v102_seam_orientation.py	One orientation: the anchor is the <i>stationary repeller</i> in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$, $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate Δ constant; SdS entropy has Nariai as its <i>unique</i> stationary point, curvature $\frac{2}{9}= \mathbb{Z}_2 /N_{\text{fam}}^2$; evaporation = entropy ascent away from the anchor [I]; “one variational principle of the seam” stays a reading [P]
v103_trisection_normal_form.py	Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow \cos 3\theta=-3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m=\cos \psi/N_{\text{fam}}$ and $S_{\text{tot}}/S_{\text{dS}}=\frac{4}{3}-\frac{2}{3}\cos \frac{2\psi}{3}$ — one cosine of glue atoms; canonical curvature $(\frac{2}{3})^3=(\mathbb{Z}_2 /N_{\text{fam}})^{N_{\text{fam}}}$ (Koide constant to the family power); invariant slope $d\sigma/dm _N=-\frac{8}{9}=-\text{rank } E_8/N_{\text{fam}}^2$; bridge: flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, missing = one <i>clock</i> [I] + [P]
v104_nariai_clock.py	The <i>classical</i> Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the dS_2 static-patch equation with $m^2=-2\Lambda=- \mathbb{Z}_2 \Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\text{clock}}=(\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1, -2\}$; Nariai cubic = $(t-1) \cdot \chi_{\text{clock}}$; entropy-deviation rate $2H= \mathbb{Z}_2 H$; honest $(2/3)$ -test <i>negative</i> for the classical clock — the quantum clock is the one remaining [P] of the seam variational principle [I] + [P]
v105_residual_inventory.py	The residual inventory: <i>one constant</i> $(2/3= \mathbb{Z}_2 /N_{\text{fam}}$ exact in seven places across both sectors), <i>one anchor</i> (three dynamical appearances: configuration, Nariai roots, clock spectrum), <i>relocation theorem</i> (cosmological quantum clock suppressed by the Λ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10 \Rightarrow$ the clock must be the seam transfer operator — one [P] identification), and the <i>complete machine-pinned gap list</i> : exactly five structural objects + one scale + π ; a sixth gap would fail the script [I] + [P]
v106_review_validation.py	External-review validation: seed normal form $\varphi_0^{\text{ret}}=\frac{ \mu_4 }{N_{\text{fam}}}c_3+\Omega_{\text{adm}}c_3^{ \mu_4 }$ exact [I]; hypercharge moment $\text{Tr } X^2=120=5!$ computed from the charge table; factorial spine $5!= R^+(E_8) =\sum \text{exponents}$, $240=2 \cdot 120$, $1920=2^4 \cdot 5!$; degree-2 inventory (Pascal $K=2$ unique at $g=5$, glue norms sum 2, $c_3=\frac{1}{2} \times \frac{1}{4\pi}$, $\binom{5}{2}=10$); named [P] hypothesis <i>Quadratic Boundary Locality</i> ($\Rightarrow K=2$ forced; would upgrade the Pascal selection to [L]); audit $240=16 \times 15$ non-unique [I] + [P]
v107_quantum_clock_target.py	Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0, -1, -2\} = -\text{Spec}(V) = -N_{\text{fam}} \times \text{cusp}$ (the transfer operator’s own grading) [I]; exact target identity $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi) \cdot \Lambda/\bar{M}_{\text{Pl}}^2$: 10^{-122} cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [I]; identification stays [P]
v108_pascal_ladder.py	Pascal Ladder Theorem: $2^{g-1}=\sum_{k \leq K} \binom{g}{k} \Leftrightarrow g=2K+1$ (odd- g midpoint identity + even- g straddle) $\Rightarrow$ carrier selection $g=5$ <i>exactly equivalent</i> to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\text{fam}}=8$ (v14 overdetermination) [I]/[L] + [P]
v109_sheet_pairing.py	Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm \frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+ \otimes S^+ = 0$; odd g flips parity); $S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4$ exactly ($256=1+45+210$, zero-modes $(1, 5, 10)$ = the code); sheet-diagonal = odd forms $(2\Lambda^1+2\Lambda^3+\Lambda^5)$; tower tops at Λ^{2K} , $K=2 \Rightarrow$ the seam scalar kernel is necessarily a <i>sheet pairing</i> [I] + [P]

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Script	What it machine-checks (<i>continued</i>)
v110_calderon_sheet.py	Calderón-sheet selection theorem: a Calderón involution ε on $S^+ \oplus S^-$ certifies a scalar two-point datum <i>iff</i> it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2= \mathbb{Z}_2 $ invariant kernels = the two Lagrangian glues; ladder genericity ($g=3, 5, 7$): sheet-oddness forces $K=(g-1)/2$, <i>not</i> $g=5$ (anti-overclaim); QBL residue split into <i>interface</i> (“ ε sheet-odd”, natural from the one-sided collar) + <i>analytic core</i> (slot-degrees ≤ 2) [I] + [P]
v111_quadratic_transport.py	Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics (= $\mathfrak{so}(10)$, the Λ^2 term of $1+45+210$) sheet-even, generate the code from the vacuum <i>and</i> span the full $\text{End}(S^+)=256$ at word length 2 $\Rightarrow$ <i>transport degree exactly 2</i> (degree ≤ 1 generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank [I] + [P]
v112_selfcounting_channel.py	Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so $\#\text{kernels} = 2^{g-1}$ <i>exactly</i> ; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1} = \sum_{m \leq K} \binom{g}{m}$ is <i>two countings of one set</i> (an identity, not a requirement); Hodge fold $\binom{g}{2j} = \binom{g}{\min(2j, g-2j)}$; QBL residue = one input (“a single scalar 2-point kernel”), selection carried by rank-8+integrality [I] + [P]
v113_quasifree_kernel.py	One kernel is the whole net: carrier $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1 = 16$ free Majoranas, $c=5+3=8$ at every tower level (only the glue grows, index $2 \times 2 = 4 = \mu_4 $); Wick/Pfaffian exact (all $210+210$ higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\text{car}}$, seam level $8 = \text{rank } E_8$ ($c = \text{rank of the kernel}$); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [L] [I] + [P]
v114_torsion_delta.py	Torsion normal form + $\delta = \frac{1}{2}$ theorem: flatness of the $(U_{\text{wall}}) \mathbb{Z}_4$ -family $\Leftrightarrow (MU)^4 = \mathbf{1}$ ($\mu_4 = \text{order of the twisted cusp generator}$); on the involutive branch ($T^2 = \mathbf{1}$) the cusp trace forces $\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2})$ with $\text{spec } \{1, \omega, \omega^2\}$ automatic $\Rightarrow \delta = \frac{1}{2}$ exact and constant (v20’s hand value = torsion identity); census: $\{1, i, -i\}$ branch varies, v33 point in its $d_1=0$ slice; residue = bundle-side branch selection [I] + [N] + [P]
v115_anchor_residue.py	The anchor pins the residue: μ_4 -average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow \text{diag } A_0 = a/ \mu_4 $; off-weights uniquely $(8, 0, 5)/144$ ($= (\text{rank } E_8, 0, g_{\text{car}})/(\mu_4 N_{\text{fam}})^2$, total $13 = \Delta_Q$); <i>exact</i> residue A_0^* with $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ is the RH solution ($\ M_\infty - \mathbf{1}\ \sim 10^{-10}$); anchor forces $d_1=0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [I] + [N]
v116_resonance_uniqueness.py	Resonance theorem (v115 [N] $\rightarrow$ [I] , linear — no Gröbner): twisted μ_4 averages collapse to $B_1 = 4a_{12}E_{12} + 4\overline{a}_{13}E_{31}$; anchor exponents $\text{diag}(2, 1, 1)$ give resonance gap 1 with obstruction $4\overline{a}_{13} \Rightarrow M_\infty = \mathbf{1} \Leftrightarrow a_{13}=0$; + $(8, 0, 5)/144$ + Diagonaleichung $\Rightarrow$ flat anchor locus = <i>one</i> gauge orbit = exact A_0^* — U_{wall} bundle datum algebraically unique; sharpness control $\ M_\infty - \mathbf{1}\ \sim 8.5 a_{13} $ [I] + [N]
v117_monodromy_weyl_a3.py	The monodromy is $W(A_3)$: the unitarised holonomy of A_0^* is <i>exact</i> (entries in $\frac{1}{2}\mathbb{Z}[i]$), $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, \frac{i}{2}) \Rightarrow d_1=0, \delta = \frac{1}{2}$ are matrix entries; $\langle U, \widetilde{M}_0 \rangle$ enumerated exactly: order 24, statistics $(1, 9, 8, 6)$, characters $(3, -1, 0, 1) = S_4 = W(A_3)$ (standard $\otimes$ sign) — the flavor wall realises the A_3 glue factor as its Weyl group; ODE match 3.5×10^{-10} [I] + [N]
v118_hexagon_family_dictionary.py	First exact H2 piece: $\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$); $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 \Rightarrow \det_{\mp} = (\frac{7}{8}, \frac{9}{8})$; lepton coefficients = resolvent determinants ($c_e = \mu_4 /(2 \det_-)$, $c_\mu = N_{\text{fam}}/(2 \det_+)$, $c_\tau = \mu_4 \det_- / N_{\text{fam}}$, product = $ \mu_4 /\det_+ = \frac{32}{9}$); $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48 = \Omega_{\text{adm}}$; the address table stays open [I] + [P]

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Script	What it machine-checks (<i>continued</i>)
v119_review_validation_2.py	Second review validation: anchor ratio triad $(e_3, e_1, e_2)/p_0 = (\frac{2}{3}, \frac{4}{3}, \frac{5}{3}) = (\text{Koide branch, seed gain, carrier branch; flow critical points } (2, 5) = (e_3, e_2));$ the 121 audit lemma $(\mathbf{1}+a)^T R(\mathbf{1}+a) = 121 = 11^2 = \ \text{Pl}(K)\ _1^2$ (orderings $\{105, 121, 135\}$, only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\text{adm}} = 2p_1p_2$, $10b_1 = 1 + p_1p_3$, $\Delta_Y = e_2^2 = p_0^2 + 2 \text{ rank}$; flavor surface = v94/v95 (presentational) [I]
v120_address_table.py	Address table (frozen v18/v20 integers): lepton words $(8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}})$; addresses = divmod by the hexagon ($p_2 = 6$): $e \rightarrow (2, 1)$, $\mu \rightarrow (5, 0)$, $\tau \rightarrow (3, 0)$, $\sum r = p_3$; sheet parity: τ sits at the real twisted eigenvalue -1 (= the product-rule site); quark sums: up = p_3 , down = $p_1 + p_3$, all quarks = $24 = W(A_3) $, all nine = $40 = p_1p_3 = a^T Ra$; top at the vacuum site $(0, 0)$; the assignment derivation stays open [I] + [P]
v121_address_pinning.py	Address pinning: $L = R + 2U$ (v95) $\Rightarrow$ the word table = the residue operator R + the generation-1 winding, R column 1 = the anchor a ; margins = atoms (rows (e_1, rank, p_3) , columns $(e_1, \Delta_Q, g_{\text{car}})$); <i>census theorem</i> : among all hexagon matrices with atom margins (17 of them) exactly <i>one</i> has $\det = 8$ — and it is R (also unique with SNF $(1, 1, 8)$; all 17 determinants distinct) $\Rightarrow$ the address table carries <i>zero</i> residual information; the H2 residue = the six atom margins [I] + [P]
v122_margin_theorem.py	Margin theorem: the <i>established</i> selectors — the D_4 annihilator $n = (5, -9, 6)$ with $n^T R = (8, 0, 0)$ (v94), $\det R = 8$, $\det K = 4$ (the v71 quartet) — pin R <i>uniquely</i> among hexagon matrices ($64 \rightarrow 12 \rightarrow 1$) $\Rightarrow$ the atom margins, the anchor column and all sum rules are <i>consequences</i> (their input status is revoked); bonus: $\det L = 20$ holds automatically on the locus (the quartet is redundant); H2 introduces <i>no</i> new residual class [I] + [P]
v123_inventory_update.py	Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries <i>three</i> loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to <i>five</i> classes — R1 clock, R2 index-4 net, R3 EH, R4' selector establishment (replacing the <i>discharged</i> H2 dictionary), R5 Q realisation — + v_{geo} , π ; a sixth class fails the script [I]
v124_resummed_clock.py	Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1 - n/N_{\text{fam}})^{p_2} \Rightarrow \text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ reproduces $\{0, \Delta, 6 \ln 3\}$, the bend $\log_{3/2} 3$ in one line; the linearisation carries slope $ \mathbb{Z}_2 = 2$ relative to the classical GP clock $\{0, -1, -2\}$; the pole at $n = N_{\text{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [I] + [P]
v125_glue_qsystem.py	Glue Q -system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q -system axioms exactly (associativity, unit, Frobenius, specialness $mm^* = \mathbb{Z}_4 \mathbf{1} \Rightarrow$ Jones index $4 = \mu_4 $ (the v89 value); locality = isotropy ($q(k(1, 1)) = k^2 \in \mathbb{Z}$, v92); halfway sub- Q -system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ step, $4 = 2 \times 2$; R2 sharpens from “find” to “identify” [I] + [P]
v126_clock_wall_bridge.py	Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}} = \{0, \frac{1}{3}, \frac{2}{3}\}$ = exactly $\text{spec } A_0^*$ (the parabolic MS weights); $\lambda_n = (1 - \alpha_n)^{p_2}$; checkpoint 1 passed <i>classically</i> ($p_2/N_{\text{fam}} = 2 = \mathbb{Z}_2 $ = the v104 entropy rate $2H$); $\det(\mathbf{1} - A_0^*) = \frac{9}{2}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [I] + [P]
v127_ring_resummation.py	The clock is a log determinant: $\text{rate} = -p_2 \text{Tr} \ln(\mathbf{1} - \alpha P)$ — six identical ring (RPA) towers, <i>one per hexagon site</i> , coupling = the parabolic weight; the $k=1$ ring = the classical term, the $k=2$ ring = the first quantum correction; $\kappa_{\text{seam}} = \frac{1}{3\pi} = \alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha \rightarrow 1$; v107's “O(1) resummation” is thereby <i>typed</i> (ring/log-det), its derivation open [I] + [P]
v128_graded_hull.py	Graded hull + zero-mode multiplicity: the 240 E_8 roots explicit in $D_5 \oplus A_3$ + glue coordinates, coset decomposition $240 = 52 + 64 + 60 + 64$; the grading is exactly additive on <i>all</i> 6720 root-pair sums $\Rightarrow E_8$ is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue = Q -system); ring multiplicity $6 = 2(2l+1) _{l=1}$ = the sheet-doubled GP zero modes [I] + [P]

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Script	What it machine-checks (<i>continued</i>)
v129_entropy_power_ law.py	The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (v101); rate = $p_2 \ln(S_{dS}/S)$; triple identity $\alpha=n/N_{\text{fam}}=$ MS weight = entropy-deficit share $\Rightarrow$ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma \propto (S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) [I] + [P]
v130_born_ square.py	Born square: amplitude $\sim (S/S_{dS})^{n_{\text{zero}}/2}=(S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability = $ A ^2=(S/S_{dS})^6 \Rightarrow h=n_{\text{zero}}/2=3=N_{\text{fam}}$, $p_2=2h$ (the Born rule) — the v129 weight audit is <i>derived</i> ; the 3 = the dimension of the $l=1$ space = the number of proper conformal Killing vectors of S^2 ; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure [I] + [P]
v131_measure_is_ area.py	The measure is the area: $\ Y_{1m}\ ^2=r^2=A/(4\pi)$ <i>exactly</i> (all three $l=1$ modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2} \Rightarrow$ amplitude S^3 , Born S^6 — every exponent factor with a derivation origin; the 4π = the same Gauss–Bonnet primitive as in c_3 , cancelling in all ratios; the R1 residue = <i>one</i> det-ratio cancellation $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ [I] + [P]
v132_detratio_ anomaly.py	Det-ratio anomaly = the Koide constant: $\zeta(0) _{\det'}$ of the GP sphere operator $-\Delta-2$ (the 3 zero modes removed) $= a_1-3 = \frac{7}{3}-3 = -\frac{2}{3} = - \mathbb{Z}_2 /N_{\text{fam}}$ exactly ($a_1 = \frac{1}{N_{\text{fam}}} + \mathbb{Z}_2 $; numerical continuation $\sim 10^{-7}$) — the eighth $2/3$ appearance, the first as a <i>spectral</i> anomaly; consequence $\det'(cL)=c^{\zeta(0)} \det'(L)$; the R1 residue = <i>one</i> $\zeta(0)$ budget statement (the 4d sum) [I] + [P]
v133_zeta_ budget.py	The ζ budget, both routes exact (Euclidean Nariai = $S^2 \times S^2$): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3} = -e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2 = \frac{161}{45}$, $\zeta(0) = -\frac{109}{45}$ (<i>no</i> atom; continuation $\sim 10^{-8}$); zero modes $3+3$ = the horizon pair $\Rightarrow$ the budget structurally favours the <i>reduced</i> seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) [I] + [P]
v134_dual_ anchor.py	Dual anchor lemma: $d:=a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$, $d \cdot \mathbf{1}=0$, $d \cdot a=1$, $(1, 1, -2)=-2d$ — the inverse flavor response <i>is</i> the Nariai root (v101); the invariance is exact via Sherman–Morrison ($\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1}=\frac{1}{4}(1, 1, -1)$; controls: $\mathbf{1}, e_1, n$ do <i>not</i> lie on the plane); (d, n) = the dual normal pair of the flavor boundary; a third purely <i>algebraic</i> leg of the flavor $\leftrightarrow$ horizon bridge [I] + [P]
v135_det_ surface.py	Determinant surface: $\det M(s, t)=3s(t+1)(t+2)+t^2+5t+8$ — two s -independent iso-volume walls $t=-2 \Rightarrow 2= \mathbb{Z}_2 $, $t=-1 \Rightarrow 4= \mu_4 $ ($\det K=4$ is a <i>layer</i>); winding line $6s+8 \Rightarrow (8, 14, 20)$, $\det F=32=2^{g_{\text{car}}}$; $\det B(s, 0)=2 \det M(s, 0)$ exactly, B -wall only at $t=-2$; row-sum axes $C=(7, 11, 13)$, $U=(3, 3, 3)$, $V=(1, 2, 3)=\text{Spec}(Q_+)$; micro-identities $e_1^2+e_2^2=41=10b_1$, $p_1^2+p_2^2=52$ = the carrier coset root count (v128) [I] + [P]
v136_dual_normal_ selector.py	Dual-normal selector: (d, n) pins R <i>columnwise</i> — $d \cdot c_j = a_j$, $n \cdot c_j = 8\delta_{1j}$, kernel = primitive $(6, 8, 7)$, census: each column unique (1 of 216) $\Rightarrow R4'$ shrinks from 6^9 to three column problems; the A_0^* zero mode carries $\sigma=(2, -9, 5)=(\mathbb{Z}_2 , -N_{\text{fam}}^2, g_{\text{car}})$ ($\ v\ ^2=\frac{16}{9}$, $16=\dim S^+$; $\text{adj}=\frac{2}{9}P_v$ = the SdS curvature); $n \cdot \sigma=121=11^2$; $W(A_3)$ orbit control: the naive lift $\sigma \rightarrow n$ fails — a genuine mechanism, not a group element [I] + [P]
v137_qplus_ cohomology.py	The Q_+ grading is cohomology: $H^1(\mathbb{P} \setminus \mu_4)=\chi_1 \oplus \chi_2 \oplus \chi_3$ ($\omega_k=z^{k-1}dz/(z^4-1)$, character i^k , residues $\zeta^k/4$); degrees $(1, 2, 3)=\text{Spec}(Q_+)$ = the A_3 exponents ($h=4= \mu_4 $); cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}=\text{spec}(A_0^*) \Rightarrow$ MS weights = cusp classes = characters = Q_+ : <i>one</i> spectrum, four readouts; the QGEO residue = the naturality of the canonical map [I] + [P]

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Script	What it machine-checks (<i>continued</i>)
v138_vw_ firewall.py	Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\text{PI}} = -\frac{98}{45} = \frac{22}{45} - \frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); edge match : $\alpha_{\text{edge}} = -\frac{8}{3} = 2 \times (-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom $\Rightarrow$ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall [I] + [P]
v139_selector_ triangle.py	Selector triangle: $d = \frac{3}{2}a - 2 \cdot 1$ exactly (pure anchor data, no R) $\Rightarrow$ the first selector is <i>derived</i> ; the frame $(1, a, \sigma)$ has $\det = 11 = \ \text{Pl}(K)\ _1$, and n is the <i>unique</i> covector with atom pairings $(2, 8, 121) = (\mathbb{Z}_2 , \text{rank } E_8, 11^2)$; on the constrained line the pairing is $11(11+t)$, a square only at the true n ; review lift exact (audit): $n = \text{reverse}(\sigma) + \mu_4 e_3$; $R4'$ residue = three pairings [I] + [P]
v140_canonical_ map.py	The canonical map exists and is rigid: the plain deck U (characters $(0, 1, 3)$) cannot match H^1 ; the <i>sheet-twisted</i> $-U$ $((2, 3, 1))$ and the cusp exponential i^{Q+} $((1, 2, 3))$ both carry the H^1 character set $\{1, 2, 3\}$ (the v118 sign twist); multiplicity-freeness $\Rightarrow$ Schur-rigid equivariant isomorphism (unique up to scalars); i^{Q+} pulls the Euler degree back to Q_+ , $-U$ to $Q_+ \circ \rho$ — the QGEO residue collapses to <i>one</i> $\mathbb{Z}_3$ choice [I] + [P]
v100_numerology_ null_mc.py	Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$, robust to budget/window variation; $2 \times 2 \cdot 10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly <i>one</i> hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$, <i>conditional on the declared grammar</i> [N]

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim $\leftrightarrow$ script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all six documents are written against). The [F] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).